

BALL MILL 5

OPERATING MANUAL

Fictional Technical Documentation — Prototype Demonstration

Document No.	BM05-OM-001
Equipment Tag	BM-05
Equipment	Ball Mill 5
Application	Cement Grinding
Model	FPM-BM5200-2C (Fictional)
Manufacturer	Fictitious Process Machinery Ltd.
Revision	Rev. 01
Issue Date	01 September 2026
Document Status	Sample / Non-Operational

IMPORTANT: This document is entirely fictional and has been prepared for software prototype and demonstration purposes. It is not an actual manufacturer manual and must not be used for operating or maintaining real equipment.

1. DOCUMENT CONTROL

Revision	Date	Description	Prepared By	Approved By
00	15 Aug 2026	Initial draft	Engineering Systems	Prototype Review
01	01 Sep 2026	Prototype issue	Engineering Systems	Prototype Review

1.1 Purpose

This manual provides fictional operating information for Ball Mill 5 (BM-05), a two-chamber horizontal ball mill used in the grinding of cementitious material. It is structured to resemble an industrial equipment operating manual and is intended solely to demonstrate document management, search and viewing functions in a maintenance information system.

1.2 Equipment Description

BM-05 is a fictional two-chamber grinding mill consisting of a fabricated cylindrical shell, inlet and outlet assemblies, first- and second-chamber liners, an intermediate diaphragm, a main drive system, lubrication equipment, monitoring instrumentation and local/remote control interfaces.

1.3 Intended Operating Philosophy

The mill is designed for continuous operation within defined process and mechanical limits. The operator is expected to verify permissives, lubrication availability, auxiliary equipment status and material-flow conditions before starting the main drive.

1.4 Related Documents

Document	Reference
Maintenance Manual	BM05-MM-001
Electrical Drawing	BM05-EL-001
Troubleshooting Guide	BM05-TG-001
Daily Inspection Checklist	BM05-CK-001

2. TECHNICAL DESCRIPTION

2.1 Principal Data

Parameter	Fictional Design Value
Mill Type	Horizontal, two-chamber ball mill
Shell Inside Diameter	5.20 m
Effective Grinding Length	12.80 m
Chamber 1 Length	4.20 m
Chamber 2 Length	8.60 m
Nominal Mill Speed	14.2 rpm
Maximum Permitted Speed	16.0 rpm
Main Drive Motor	4,500 kW, 6.6 kV
Gearbox Ratio	Approx. 70:1
Lubrication	Central forced-oil system
Control System	PLC-based supervisory control
Mill Rotation	Clockwise when viewed from drive end

2.2 Main Assemblies

Item	Assembly	Function
01	Mill Shell	Contains grinding media and material being processed.
02	First Chamber Liners	Protect shell and promote coarse grinding action.
03	Intermediate Diaphragm	Controls material transfer between chambers.
04	Second Chamber Liners	Provide fine grinding environment.
05	Main Bearings	Support the rotating mill shell.
06	Drive System	Transfers motor power to the mill.
07	Oil Lubrication Unit	Supplies filtered and cooled oil to critical components.

2.3 Key Instruments

Fictional instrumentation includes bearing temperature sensors, gearbox oil-pressure and temperature transmitters, mill motor current measurement, vibration monitoring, lubrication flow switches, and emergency-stop devices.

3. SAFETY AND OPERATING LIMITS

WARNING: Real industrial ball mills contain stored mechanical energy, rotating components, high-voltage electrical equipment, heavy components and other hazards. The procedures in this fictional document are not a substitute for site-specific risk assessment, isolation procedures, permits or manufacturer instructions.

3.1 General Safety Rules

- Only trained and authorized personnel may operate or access the equipment.
- All guards and covers must be installed before normal operation.
- Emergency-stop devices must remain accessible and functional.
- No person may enter the mill, drive train or guarded area while the equipment can start or rotate.
- Electrical and mechanical isolation must be verified before maintenance or internal inspection.
- Operators must remain within approved operating limits and report abnormal conditions immediately.

3.2 Fictional Operating Limits

Condition	Normal Range	High Alarm	Trip / Shutdown
Main bearing temperature	35–65 °C	70 °C	80 °C
Gearbox oil temperature	35–55 °C	60 °C	70 °C
Gearbox oil pressure	2.0–3.5 bar	—	<1.5 bar
Main motor current	60–85% FLA	90% FLA	100% FLA
Mill vibration	≤ 4.0 mm/s	5.5 mm/s	7.0 mm/s

3.3 Abnormal Conditions

Operators should stop the mill when there is evidence of severe mechanical noise, rapidly increasing bearing temperature, loss of lubrication pressure, excessive vibration, electrical fault, fire, structural damage or any other condition that presents an immediate hazard.

4. PRE-START CHECKS

The following fictional checklist represents the minimum checks expected before starting BM-05. Actual facilities should use approved site procedures.

No.	Check	Acceptance Condition
01	Mill area	Clear of personnel, tools and loose materials.
02	Guards	All drive and rotating-part guards secured.
03	Main bearings	Oil level/flow available and no visible leakage.
04	Gearbox	Oil level normal; no abnormal leakage.
05	Lubrication unit	Pump available; pressure permissive healthy.
06	Cooling system	Cooling water/air available as applicable.
07	Material circuit	Downstream equipment available.
08	Feed system	Ready and free from abnormal blockage.
09	Instrumentation	Required measurements available and plausible.
10	Electrical system	No active trip or emergency-stop condition.
11	Communication	Operator and field personnel communication confirmed.
12	PLC permissives	Start permissive displayed as healthy.

4.1 Pre-Start Sequence

1. Inspect the mill area. 2. Confirm personnel are clear. 3. Start the lubrication system. 4. Confirm oil pressure and flow permissives. 5. Start required auxiliary equipment. 6. Confirm downstream process availability. 7. Confirm the mill start permissive. 8. Start the main drive from the approved control station.

5. STARTING AND NORMAL OPERATION

5.1 Starting Procedure

Step	Operator Action	Expected Response
1	Confirm all pre-start checks complete.	Start permissive available.
2	Start lubrication system.	Pump runs; pressure rises into normal range.
3	Confirm auxiliaries.	Cooling and supporting systems available.
4	Start main motor.	Motor accelerates without abnormal alarm.
5	Observe bearing and gearbox values.	Temperatures remain stable.
6	Introduce feed gradually.	Material flow establishes smoothly.
7	Stabilize process.	Motor load and product flow reach target.

5.2 Normal Monitoring

During operation, the operator should periodically observe motor load, mill bearing temperatures, gearbox oil pressure and temperature, vibration, lubrication status, feed rate and product-flow conditions. Trends are generally more useful than isolated readings.

5.3 Normal Operating Observations

- No persistent metallic impact or unusual mechanical noise.
- Bearing temperatures rise gradually and stabilize.
- Lubrication pressure remains within the permitted range.
- Motor current follows changes in feed and process conditions without sustained overload.
- Vibration remains stable and below the fictional alarm threshold.
- No active protection trips or critical interlock alarms.

5.4 Operator Log

Time	Motor Load	Bearing Temp.	Oil Pressure	Vibration	Remarks
06:00	—	—	—	—	Sample entry
10:00	—	—	—	—	Sample entry
14:00	—	—	—	—	Sample entry
18:00	—	—	—	—	Sample entry

6. STOPPING PROCEDURE

6.1 Normal Stop

For a normal stop, reduce or stop material feed according to the process sequence, allow the mill to clear as required by the operating philosophy, and stop the main drive from the approved control station. Keep the lubrication system operating for the required post-run period before shutting it down.

Step	Action	Purpose
1	Reduce/stop feed.	Prevent unnecessary material accumulation.
2	Allow process to stabilize.	Reduce mill load before stopping.
3	Stop main drive.	Remove rotation.
4	Maintain lubrication.	Protect bearings and gearbox during coast-down.
5	Monitor temperatures.	Confirm controlled shutdown.
6	Stop lubrication after approved delay.	Complete shutdown sequence.

6.2 Emergency Stop

If an immediate hazardous condition occurs, use the designated emergency-stop system in accordance with site procedures. After an emergency stop, the cause must be identified and the equipment must not be restarted until the responsible authorized personnel have determined that it is safe to do so.

6.3 Restart After Trip

Record the active alarm and trip condition, inspect the relevant equipment, verify that the initiating condition has cleared, and follow the approved reset and restart sequence. Repeated trips must not be bypassed merely to restore production.

6.4 Shutdown Checklist

Check	Status
Feed stopped	■
Main drive stopped	■
Lubrication post-run completed	■
No abnormal temperature rise	■
Area left in safe condition	■

7. CONTROL SYSTEM AND INTERLOCKS

7.1 Simplified Control Philosophy

The fictional BM-05 control system uses a PLC to monitor permissives, process signals and protective devices. A start command is accepted only when the required permissive chain is healthy.

7.2 Representative Start Permissives

Permissive	Signal	Required State
Emergency stop	E-STOP-OK	Healthy
Lubrication pressure	PS-05	Above minimum
Oil temperature	TS-05	Within start range
Main bearing temperature	TT-05A/B	Below start limit
Gearbox ready	GB-READY	Healthy
Downstream equipment	DS-READY	Running/ready
Motor protection	MTR-OK	Healthy
PLC system	PLC-HEALTHY	Healthy

7.3 Representative Protective Trips

Trip Code	Description	Typical Response
BM05-T01	Low lubrication pressure	Stop main drive / alarm
BM05-T02	High bearing temperature	Controlled shutdown / alarm
BM05-T03	High vibration	Trip drive / investigate
BM05-T04	Motor overload	Electrical protection trip
BM05-T05	Emergency stop	Immediate drive shutdown
BM05-T06	Gearbox oil high temperature	Alarm or trip depending on level

7.4 Alarm Handling

Alarm messages should be acknowledged only after the operator understands the condition. Alarm acknowledgement does not mean the fault has been corrected.

8. ROUTINE OPERATOR INSPECTION

8.1 Daily Inspection

Inspection Point	Method	Acceptance
Main bearing housings	Visual / temperature display	No leakage; stable temperature.
Gearbox	Visual / HMI	No abnormal noise, leakage or temperature.
Lubrication unit	Gauge / sight glass	Pressure and level normal.
Mill shell	Visual	No visible damage or abnormal leakage.
Drive coupling	Visual from safe location	Guard intact; no abnormal movement.
Motor	HMI / visual	Load stable; no abnormal noise.
Vibration	Monitoring system	Within normal trend.
Feed chute	Visual from safe location	No uncontrolled blockage or spillage.

8.2 Weekly Checks

- Review temperature and vibration trends for abnormal changes.
- Inspect lubrication system filters, indicators and visible pipework.
- Check for new oil leaks around bearing and gearbox areas.
- Review alarm history for repeated or intermittent events.
- Verify that guards, warning signs and housekeeping remain satisfactory.

8.3 Operator Records

All abnormal observations should be entered into the maintenance or shift reporting system. A useful record should include the equipment tag, date/time, observed condition, relevant readings, action taken and person responsible.

Prototype note: The values and procedures in this manual are intentionally fictional and are provided only to populate the demonstration document repository.

9. QUICK REFERENCE & DOCUMENT INDEX

9.1 Quick Start Reference

Sequence	Confirm
1	Area clear and personnel accounted for
2	Guards and safety devices healthy
3	Lubrication system running
4	Oil pressure/flow permissive healthy
5	Auxiliary systems ready
6	Downstream process available
7	PLC start permissive available
8	Start main drive
9	Monitor during acceleration
10	Introduce feed gradually

9.2 Document Index

Document	Number	Purpose
Operating Manual	BM05-OM-001	Operation and operator checks
Maintenance Manual	BM05-MM-001	Maintenance and inspection guidance
Electrical Drawing	BM05-EL-001	Fictional electrical/control reference
Troubleshooting Guide	BM05-TG-001	Fault diagnosis and corrective guidance
Inspection Checklist	BM05-CK-001	Routine inspection records

9.3 Prototype Disclaimer

This 10-page document is a fictional sample created for demonstration of a digital equipment-documentation platform. Equipment dimensions, ratings, trip values, document numbers, manufacturer information, procedures and technical parameters are invented. No real company manual, manufacturer drawing or proprietary technical document has been reproduced.

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